selected as primary. The primary and the other source(s) should be indicated as such.

7.4.3 Optional functions

Optionally, the following information may be displayed:

- true scaled ship symbols and CPA/TCPA and bow crossing range (BCR) / bow crossing time (BCT) related to the real dimensions

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SOURCE: MSC.252(83).pdf
7.4.3 Optional functions

Optionally, the following information may be displayed:

- true scaled ship symbols and CPA/TCPA and bow crossing range (BCR) / bow crossing time (BCT) related to the real dimensions
- chart data from the common database of INS: traffic-related object layers.

7.5 Task “Navigation Control Data”

7.5.1 General

To support the manual and automatic control of th e ship’s primary movement the INS navigation control task should provide the following functionality:

- display of data for the manual control of the ship’s primary movement
- display of data for the automatic control of the ship’s primary movement
- presentation and handling of external safety related messages.

7.5.2 Presentation of navigation c ontrol data for manual control

7.5.2.1 For manual control of the ship’s primary movement the INS navigation control display should allow at least to display the following information:

- under keel clearance (UKC) and UKC profile
- STW, SOG, COG
- position
- heading, ROT (measured or derived from change of heading)
- rudder angle
- propulsion data

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SOURCE: MSC.252(83).pdf
should allow at least to display the following information:

- under keel clearance (UKC) and UKC profile
- STW, SOG, COG
- position
- heading, ROT (measured or derived from change of heading)
- rudder angle
- propulsion data

set and drift, wind direction and speed (true and/or relative selectable by the operator), if available
the active mode of steering or speed control
time and distance to wheel-over or to the next waypoint
safety related messages e.g., AIS safety-related and binary messages, Navtex.

7.5.3 Presentation of navigation cont rol data for automatic control

7.5.3.1 For automatic control of the ship's primary movement, the INS navigation control display should allow at least and as default the display of the following information:

all information listed for manual control
set and actual radius or rate of turn to the next segment.

7.5.4 The navigation control data should be presented:

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(adopted on 8 October 2007)
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7.5.4 The navigation control data should be presented:

in digital and where appropriate in anal ogue form, e.g., mimic elements, logically arranged on and around a symbolic outline of a ship,
if applicable, together with their "set- values",
if applicable and on demand together with a history presentati on to indicate the trend of the parameter.

7.6 Task "Alert management"

7.6.1 Scope, operational requirements and alert-re lated requirements are specified in Module C of these performance standards.

7.7 Task "Status and data display"

7.7.1 Mandatory data display functions

The INS should provide the following data display functions:

presentation of mode and status information

 presentation of the ship’s static, dynamic and voyage-related AIS data
 presentation of the ship’s available relevant measured motion data together with their “set – values”
 presentation of received safety related me ssages, such as AIS safety-related and binary messages, Navtex

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SOURCE: MSC.252(83).pdf
 presentation of the ship’s available relevant measured motion data together with their “set – values”
 presentation of received safety related me ssages, such as AIS safety-related and binary messages, Navtex
 presentation of INS configuration
 presentation of sensor and source information.

7.7.2 Mandatory data management functions

The INS should provide the following management functions:

 setting of relevant parameters
 editing AIS own ship’s data and information to be transmitted by AIS messages.

7.7.3 Optional data display functions

The INS may provide on demand:

 tidal and current data
 weather data, ice data
 additional data of the tasks Navigation control and Route monitoring and AIS target data.

8 Functional requirements for INS task stations

8.1 Number of task stations

8.1.1 The number of task stations on the bridge de pends on the tasks integr ated into the INS.

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8 Functional requirements for INS task stations

8.1 Number of task stations

8.1.1 The number of task stations on the bridge de pends on the tasks integr ated into the INS.
It should support the simultaneous operation and presentation of at least the minimum set of tasks necessary to meet the carriage requirements of SOLAS regulation V/19.

8.1.2 To specify the required number of task stations the required backup arrangements as mandated by the carriage requirements of SOLAS regulation V/19 should be taken into account.

8.2 For each tasks of:

- route monitoring
- collision avoidance
- navigation control data,

a task station should be provided, if the respective task is part of the INS.

8.3 Additional tasks

For the tasks of:

- route planning,
- status and data display, and